

AI-Driven Innovations to Enhance Tea Production and Quality in Sri Lanka's Mid-Country

Nipuna Abeyssekara
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
nipunaabeysekara2001@gmail.com

Harsha Waduthanthri
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
harshawaduthanthri@gmail.com

Nimesh Bandaranayake
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka line
bandaranayakanimesh@gmail.com

Dinithi Wijerathne
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
diniwijerathne21@gmail.com

Wishalya Tissera
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
wishalya.t@slit.lk

Ishara Weerathunga
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
ishara.w@slit.lk

Abstract— The Sri Lankan tea industry faces several operational challenges, including inconsistent leaf grading, inefficient yield forecasting, unpredictable auction demand, and delayed disease/pest detection. This research presents a novel multi-modal AI framework aimed at addressing these challenges by automating key decision-making processes. The system leverages deep learning techniques for four tasks: (1) freshness grading and tea-type classification from mobile-captured leaf images, (2) yield forecasting using estate records and scanned logbooks, (3) auction demand prediction based on historical transaction data, and (4) disease/pest detection from images with seasonal risk prediction linked to climate data. Utilizing publicly available datasets and collaborating with local experts, the framework was trained on diverse data sources, ensuring applicability to Sri Lanka's unique operational conditions. The system achieved high accuracies across all tasks, demonstrating its potential as a reliable decision-support tool for farmers, tea factories, and auction stakeholders. By enabling early detection, accurate forecasting, and informed decision-making, the system offers a scalable solution to improve productivity, quality, and market competitiveness in the tea industry. This research provides a foundation for further development and clinical validation, aiming to reduce operational inefficiencies and enhance sustainability in Sri Lanka's tea sector.

Keywords—Tea Production, Deep Learning, Freshness Grading, Yield Forecasting, Auction Demand Prediction, Disease Detection, Sri Lanka, Data Digitization

I. INTRODUCTION

Sri Lanka's tea sector is a major agricultural export industry and a key source of foreign-exchange earnings [1], [2]. The sector supports rural livelihoods across plantation and smallholder systems, and widely cited assessments report that smallholders contribute a dominant share of national production [2], [1]. However, operational decision-making across estates, factories, and market-facing processes remains constrained by manual assessment and incomplete digitization, including paper-based and partially digitized workflows that limit timeliness and consistency [2], [6].

The decision-support scope as four prediction tasks aligned with practical actions from field to auction: (1)

freshness grading and tea-type or leaf-category recognition from mobile imagery [3], [1]; (2) yield digitization and forecasting using estate records and management-linked signals [4], [2]; (3) auction-oriented forecasting using historical Colombo tea auction transactions [5], [7]; and (4) field health-risk recognition from images under label scarcity and class imbalance [8], [6]. Across tasks, methods must remain robust to heterogeneous capture conditions and record formats reported in tea and agricultural digitization contexts [3], [6].

Deployment-faithful modeling is shaped by three recurring reliability constraints. First, domain shift arises across estates, devices, and illumination conditions in mobile imagery, and across scan quality, layout variation, and missing fields in historical records [3], [6]. Second, production and market targets exhibit temporal structure and non-stationarity; therefore, evaluation should preserve time ordering to avoid optimistic estimates associated with random splitting [5], [7]. Third, minority classes may carry disproportionate operational risk, requiring imbalance-aware reporting because aggregate accuracy can obscure failures on rare but consequential outcomes [8], [3].

II. LITERATURE REVIEW

Artificial intelligence (AI) in agriculture has moved from isolated prediction models to integrated decision-support systems. In tea production, AI has been applied to freshness grading, yield forecasting, auction price prediction, and plant health monitoring. Manual tea leaf grading is still widely used, but the process is subjective and inconsistent. Early automation work showed fresh-leaf sorting was feasible in controlled settings [9], and later computer-vision pipelines improved automated quality assessment [10]. Deep learning methods were then introduced for fresh leaf classification [11], but robustness drops under real field variation (lighting/background). Previous studies highlight stalk presence and coarse plucking as strong quality indicators, supporting structured feature modeling in grading systems [12].

Convolutional Neural Networks (CNNs) have improved image-based agricultural analysis by learning discriminative

color and texture patterns for tea sorting [13]. However, many studies rely on laboratory-style datasets, limiting deployment reliability in plantations. This motivates standardized mobile image capture and stronger preprocessing to reduce illumination shifts and background noise.

Yield forecasting is essential for plantation planning and factory scheduling. Statistical models such as ARIMA remain common for tea production forecasting [17], while hybrid approaches that combine simulation with machine learning improve performance under changing conditions [15]. Explainable AI has been explored to improve trust and transparency in Sri Lankan yield estimation [14]. Climatic evidence consistently shows temperature, rainfall, and humidity influence tea growth and productivity [16], reinforcing the need for climate-aware features in forecasting models.

Digitization is still a bottleneck because many estates depend on handwritten records. OCR has been widely studied for automated text extraction [18], but end-to-end integration into agricultural analytics pipelines is still limited. On the market side, tea auctions are difficult to predict due to volatility and non-stationary behavior. Seasonal models have been used for auction price forecasting [20], and machine learning methods have been used to predict weekly prices by elevation and grade [19]. Hybrid ARIMA–Random Forest variants improve stability in related volume estimation tasks [22]. Still, many demand models assume direct demand observations, while in auctions demand is often inferred from bidding dynamics rather than measured explicitly [21].

Tea plant disease detection has progressed with deep learning. CNN-based models achieve strong classification accuracy [25], and optimized architectures improve results further [24]. Mobile-based systems have been proposed for real-time disease detection and treatment guidance [23]. Yet field deployment remains difficult due to lighting variation, class imbalance, and visually similar symptoms. Invasive plant species add ecosystem risk, increasing the need for early detection under shifting climate conditions [26]. Climate–pathogen evidence indicates temperature, humidity, and rainfall affect fungal development, and climate–pathogen interaction models suggest future changes in tea-growing suitability and outbreak risk [27]. However, most work treats visual disease detection and climate-linked outbreak prediction separately. For elevation-stratified Sri Lankan plantations, integrated frameworks combining visual health detection, invasive species monitoring, and seasonal risk modeling remain limited this is the gap the proposed system targets.

III. METHODOLOGY

A. Tea Leaf Freshness Grading & Inspection

A dataset of 275 tea leaf images was collected across 66 field sessions in mid-country Sri Lankan estates using TeaVision, a custom Android application (Kotlin, Jetpack Compose, CameraX). Images were captured at 25–35 cm on a standardized white-cloth background, and labels were verified by a domain expert. Brightness normalization was applied to reduce outdoor illumination variation. Leaf segmentation used a dual pipeline: a primary on-device TFLite model and an HSV-based fallback triggered under low confidence, with morphological refinement (erosion then dilation) to improve mask quality.

For freshness grading, 25 hand-crafted features were extracted covering color (HSV/LAB statistics, greenness–browning ratios), texture (GLCM, LBP), shape (area, eccentricity, solidity, aspect ratio), and capture quality (brightness, contrast). Features were z-score normalized using a training-only StandardScaler to prevent leakage. Freshness classification was formulated as a binary task (high_quality vs. medium_quality) using an 80/20 stratified split with 5-fold cross-validation.

For spatial localization, a YOLOv8s detector was trained on seven TRI-defined classes using 148 annotated images (augmented to 726). SAHI slicing (512×512, 0.3 overlap) was used to improve small-object detection. In addition, a retrieval-augmented advisory module was built from 156 TRI documents (473 chunks), indexed with FAISS and BM25 and combined via Reciprocal Rank Fusion using Sentence-BERT embeddings. The system is deployed in a two-tier setup: TeaVision for data capture, Streamlit for visualization/detection, and FastAPI for scalable advisory serving.

B. Tea Leaf Disease, Pest, Invasive Species Detection with Treatment Guidance and Seasonal Disease Prediction

The proposed system integrates computer vision, machine learning, and natural language processing (NLP) to enable automated tea leaf disease detection, invasive species identification, seasonal disease forecasting, and farmer advisory support. The TeaLeaves dataset contains over 6,000 annotated RGB images across six classes (Healthy, Blister Blight, Brown Blight, Grey Blight, Red Rust, and Helopeltis), while the InvasiveSpecies dataset includes over 5,000 images of common invasive plants. Images were resized to 224×224 pixels, normalized to the [0,1] range, and augmented using rotation, flipping, zooming, and brightness variations. The data was split into 80% training and 20% testing sets using stratified sampling.

A custom Convolutional Neural Network (CNN) was designed for multi-class classification, consisting of convolutional layers with ReLU activation, max-pooling layers, and fully connected dense layers with Softmax output. The model was trained using the Adam optimizer with categorical cross-entropy loss, with dropout and early stopping applied to reduce overfitting. Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices, while Grad-CAM visualization was applied to highlight discriminative regions in tea leaf images.

For seasonal disease forecasting, a dataset of 15,000 environmental records containing temperature, humidity, rainfall, season, and region was used to train Random Forest and Gradient Boosting models to estimate disease risk levels. Additionally, a retrieval-based NLP advisory chatbot was integrated to provide real-time responses to farmer queries regarding diseases, treatments, and plantation management. The system was implemented using a React frontend and FastAPI backend, enabling an interactive decision-support platform for tea plantation monitoring.

C. Tea Yield Log Book Digitization and Predictive Analytics for Crop Yield Management

A total of 175 handwritten tea yield book images were collected from 3 tea estates to develop a digitization module that converts handwritten tea yield logbooks into structured

Excel datasets. Logbook images undergo perspective rectification and morphological line detection to extract individual table cells, which are then enhanced through column-specific preprocessing pipelines. A PaddleOCR recognizer based on the PP-OCRv5 architecture (DBNet detection + CRNN recognition) is fine-tuned on manually annotated logbook cell crops to adapt the model to domain-specific handwriting styles and character patterns. Post-processing suppresses low-confidence artifacts, normalizes numeric fields via regex cleaning, and resolves handwritten month names and fertilizer codes through a composite fuzzy matching scheme combining Levenshtein, Jaro-Winkler, OCR-aware edit distance, q-gram, and permterm similarity.

The second part predicts monthly yield using a SARIMAX model fitted on a field-wise panel of 5,821 field-month records, with earlier months used for training/validation and the final 12 months reserved for testing. The feature set includes lagged yield and rainfall values, short rolling aggregates of recent conditions, plucking intensity indicators, and multi-month fertilizer history as exogenous regressors.

D. Tea Market Demand Forecasting and Smart Auction Planning

A dataset of 1.5 million Colombo Tea Auction records (2023–2025), including tea grade, realized price, quantity offered, and factory attributes, was preprocessed by handling missing values and removing outliers using interquartile range (IQR) thresholding. The cleaned data were partitioned into training (80%), validation (15%), and test (5%) subsets using stratified random sampling to preserve the dataset’s temporal distribution. For auction price forecasting, ARIMA and SARIMAX time-series models were applied to capture underlying market trends and seasonality. Model performance was evaluated on the test set using mean absolute error (MAE) and root mean squared error (RMSE). To support temporal learning, lag based features and relevant auction covariates were engineered during preprocessing and implemented in Python using the statsmodels library.

A Multi-Agent System (MAS) with Buyer, Factory, and Broker agents was employed, using Proximal Policy Optimization (PPO) for the Factory and Broker agents, and Deep Q-Network (DQN) for the Buyer agent. The agents were trained to optimize actions based on auction interactions. Explainable AI (XAI) techniques, like SHAP and LIME, were incorporated to provide transparency in the agents’ decision-making.

IV. RESULT AND DISCUSSION

A. Tea Leaf Freshness Grading & Inspection

1) Binary Quality Classification

Ten machine learning models were evaluated for the binary tea leaf freshness grading task using 148 images with an 80/20 class-stratified split and 5-fold cross-validation. Among the evaluated models, the MLP neural network achieved the highest test accuracy of 86.67% with 100% training accuracy, indicating possible overfitting. Several models including AdaBoost, Logistic Regression, SVM (RBF), SVM (Linear), and Naive Bayes achieved similar test accuracies of 83.33%, showing relatively stable performance.

Random Forest achieved 80.00% accuracy, while Gradient Boosting, Decision Tree, and KNN obtained 76.67%. Although MLP produced the best test accuracy, Logistic Regression and linear SVM showed smaller train–test gaps, suggesting more reliable generalization for this relatively small dataset.

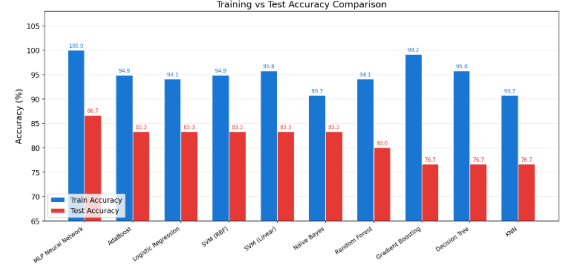


Fig. 1: Training vs Test Accuracy Comparison

2) YOLOv8 Object Detection

TABLE I: Per-Class Detection Performance(YOLOv8s)

Class	AP@0.5
Fresh_Bud_2	0.578
Coarse_pluck	0.553
Old_Leaf	0.431
Damaged_Leaf	0.213
Fresh_Bud_1	0.154
Damage_Spot	0.018
All Classes (mAP@0.5)	0.325

The YOLOv8s detector was trained on 148 annotated images, augmented to 726, across seven TRI-defined classes for 13 epochs. As shown in Table II, the model achieved an overall mAP@0.5 of 0.325. Fresh_Bud_2 and Coarse_pluck obtained the highest AP@0.5 values of 0.578 and 0.553, respectively, while Damage_Spot was the most challenging class (0.018), likely due to small object size, subtle texture variation, and limited annotations.

The training curves in Fig. 2 indicate stable convergence, with training and validation losses decreasing over epochs. Precision, recall, and mAP also improve progressively during training. The low AP for Damage_Spot suggests that small object size, subtle visual features, and limited annotations remain challenging. SAHI post-processing with 512×512 slicing and 0.3 overlap was therefore applied to improve recall for underrepresented classes.

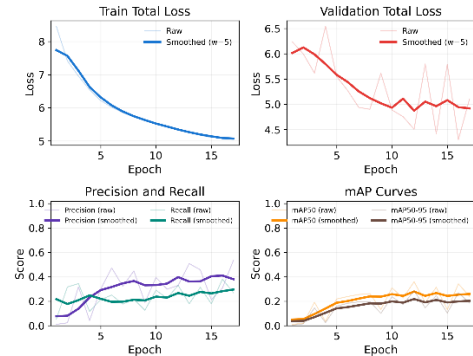


Fig. 2: Normalized confusion matrix for YOLOv8s detection across seven classes.

3) RAG Advisory Module

The retrieval pipeline was evaluated using three methods: BM25, dense embeddings (Sentence-BERT with FAISS), and a hybrid Reciprocal Rank Fusion (RRF) approach. The evaluation was conducted on a TRI knowledge corpus consisting of 156 documents segmented into 473 semantic chunks. All methods achieved near-perfect retrieval performance. BM25 obtained a hit rate, MRR, precision, and recall of 1.0 with a slightly lower nDCG of 0.98, while both dense and hybrid retrieval achieved perfect scores of 1.0 across all metrics. Although dense and hybrid approaches slightly outperformed BM25 in ranking quality, the differences were minimal due to the well-curated and domain-specific dataset. These results confirm that the retrieval module can reliably support the tea advisory system by providing relevant agronomic knowledge for decision support.

B. Tea Leaf Disease, Pest, Invasive Species Detection with Treatment Guidance and Seasonal Disease Risk Prediction

1) Leaf Disease and Pest Classification

The proposed Convolutional Neural Network (CNN) was trained using an 80:20 train-test split for 20 epochs with the Adam optimizer and categorical cross-entropy loss. The model achieved 92.49% training accuracy, 92.97% validation accuracy, and 93.90% test accuracy, indicating strong feature learning and good generalization on unseen tea leaf images. Fig. 3 illustrates the training and validation accuracy and loss curves. The steady increase in accuracy and decrease in loss demonstrate stable model convergence with minimal overfitting.

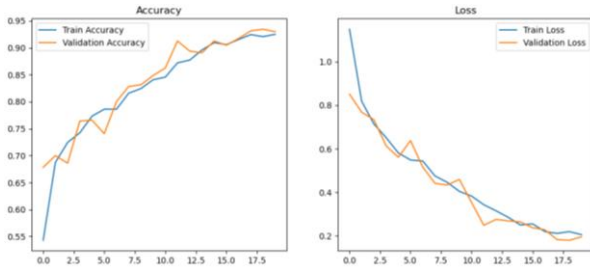


Fig. 3: Training, Validation Accuracy and Loss Curve

Fig. 4 presents the confusion matrix of the CNN model, where most predictions lie along the diagonal, confirming high classification accuracy, with only minor misclassifications among visually similar diseases.

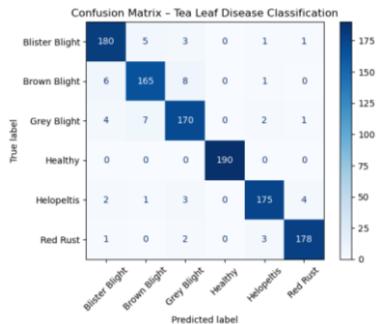


Fig. 4: Confusion Matrix for Disease CNN model

2) Invasive Species Identification

A secondary CNN model was developed to detect five invasive plant species (Lantana camara, Tridax procumbens, Mikania micrantha, Mimosa diplotricha, and Sphagneticola trilobata). The model achieved 91.49% test accuracy, demonstrating effective automated identification of invasive plants in tea plantations.

3) Seasonal Disease Prediction and NLP-Based Advisory Chatbot

A seasonal disease prediction model was developed using environmental factors such as temperature, humidity, rainfall, season, and region, achieving 73% accuracy, with strong performance for Blister Blight and Helopeltis. To support decision-making, an NLP-based advisory chatbot was integrated using a retrieval-based approach, allowing farmers to ask questions and receive immediate guidance on diseases and treatments.

TABLE II: SEASONAL MODEL PERFORMANCE METRICS

Class	Precision	Recall	F1	Support	Accuracy
Blister Blight	0.82	0.94	0.88	444	0.73
Brown Blight	0.67	0.66	0.66	618	
Grey Blight	0.69	0.61	0.65	496	
Healthy	0.73	0.54	0.62	599	
Helopeltis	0.84	0.95	0.89	412	
Red Rust	0.68	0.82	0.74	491	

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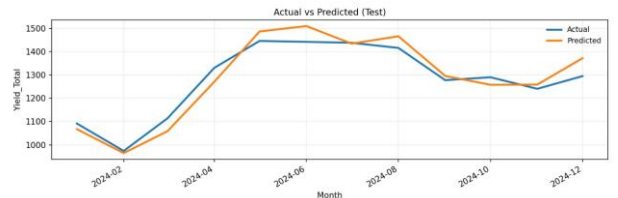


Fig. 5: Monthly Actual versus Predicted yield

D. Tea Market Demand Forecasting and Smart Auction Planning

1) Weekly Tea Price Forecasting

The forecasting framework was evaluated using three models: ARIMA, SARIMAX, and Prophet, across the three elevation categories (Low, Medium, High). Model performance was measured using MAE, RMSE, and MAPE on the test period. As summarized in TABLE V, ARIMA achieved the best overall forecasting accuracy across all elevations, reporting the lowest mean MAPE values of 4.28% (Low), 5.02% (Medium), and 5.11% (High). SARIMAX produced moderate performance, while Prophet consistently resulted in higher error values, indicating reduced suitability for weekly tea price series under the current experimental settings.

TABLE V: Overall Forecasting Performance

Elevation	Model	Mean MAE	Mean RMSE	Mean MAPE (%)
Low	ARIMA	9.14	13.15	4.28
Low	SARIMAX	15.09	23.66	7.08
Low	Prophet	29.77	52.36	15.86
Medium	ARIMA	11.23	15.45	5.02
Medium	SARIMAX	17.15	26.71	8.56
Medium	Prophet	29.61	49.93	19.33
High	ARIMA	12.08	17.39	5.11
High	SARIMAX	18.50	30.67	10.31
High	Prophet	33.68	59.91	21.05

Fig. 6 demonstrates the weekly actual vs predicted price curve for a selected grade, showing that ARIMA closely follows auction price movement during the test horizon. Fig. 7 presents the rolling 1-step-ahead forecast, confirming that rolling prediction improves stability and reduces long-horizon drift compared to static forecasting. These visualizations support the numerical results in Table I, confirming that ARIMA provides the most consistent weekly forecasting performance for the Colombo Tea Auction dataset.

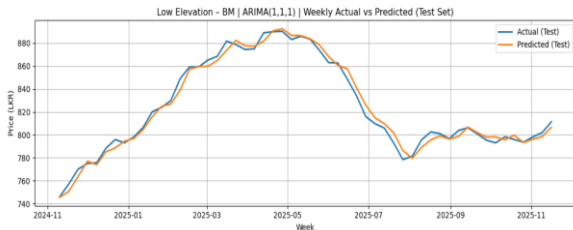


Fig. 6: Weekly Actual vs Predicted Tea Auction Prices

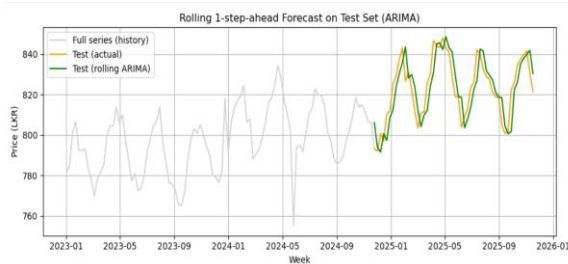


Fig. 7: Rolling 1-step-ahead Forecast (ARIMA)

2) Multi-Agent Smart Auction Planning

The proposed RL-based Multi-Agent Smart Auction Planning module was evaluated in simulation using three trained agents (Buyer-DQN, Factory-PPO, Broker-PPO) that jointly determine buyer action, factory reserve/volume strategy, and broker signal/commission per auction round. Across 100 end-to-end rounds, the MAS achieved a 96.0% clearance rate, with profitable outcomes in 63.0% of factory rounds and 96.0% of broker rounds. Under controlled rising/falling market scenarios, the Buyer and Broker achieved 100% directional decision accuracy, while broker

commissions fell within the target range in 83.5% of cases. Overall, the results indicate stable multi-agent coordination and meaningful financial behavior, with reduced profitability under falling-market conditions reflecting realistic downside risk; a live current market price input can be integrated to improve adaptability to sudden market shifts.

VI. Conclusion

This study successfully developed a multi-modal AI framework for addressing key operational challenges in the Sri Lankan tea industry, including leaf grading, yield forecasting, auction demand prediction, and disease/pest detection. The system demonstrated high accuracy in all tasks, with notable improvements in operational efficiency and predictive reliability. By leveraging deep learning and computer vision techniques, the framework provides reliable decision-support tools for stakeholders, enabling early intervention, optimized planning, and improved market forecasting. Future work will focus on expanding the system to include additional disease categories, enhancing data quality through more diverse datasets, and validating the model in real-world tea estate conditions for greater practical applicability.

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